

Rossmann Pharmaceuticals — Sales Forecasting

End-to-End Machine Learning, Deep Learning and Deployment Report

Business objective

Forecast daily sales across retail stores up to six weeks ahead so the finance team and store managers can plan promotions, staffing and inventory. The supplied project brief requires customer-behaviour EDA, store-sales prediction, ML and LSTM approaches, serialization, MLflow, DVC and a web interface.

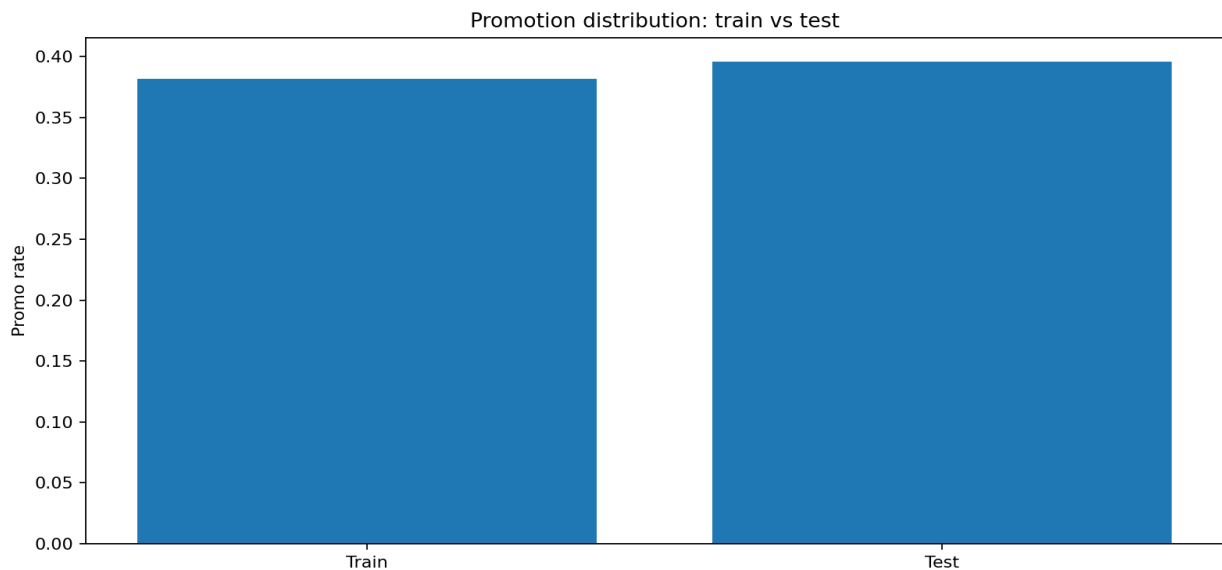
Data

The provided data contains 1,017,209 training rows covering 1115 stores from 2013-01-01 to 2015-07-31, plus 41,088 future test rows across 856 stores. Store metadata contains 1,115 stores. Sales has 16.99% zero rows.

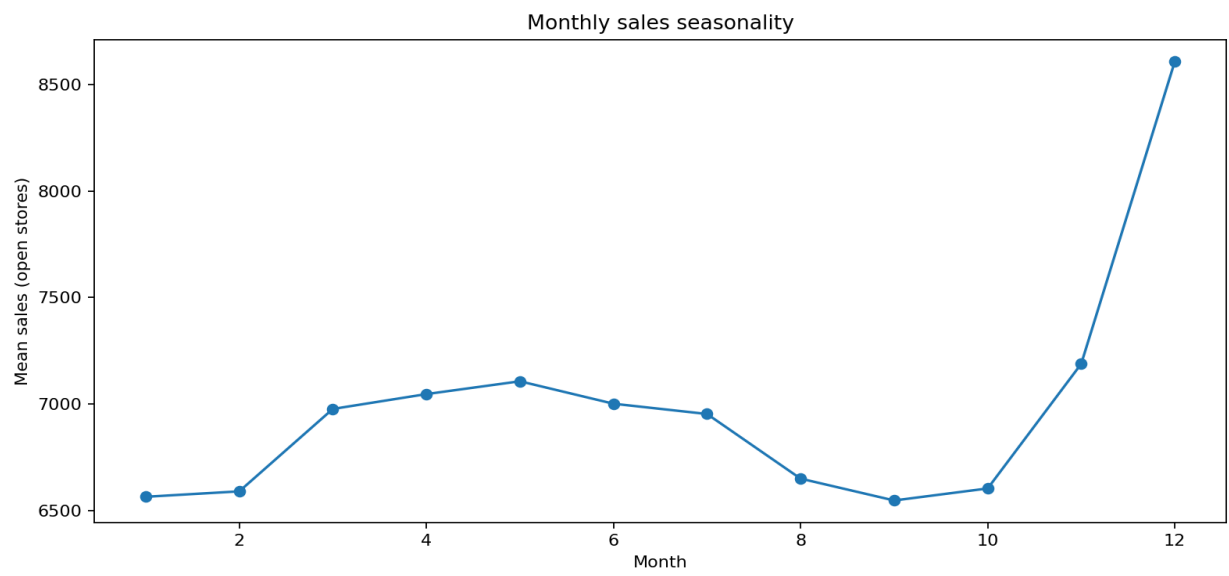
Exploratory analysis

Promo rate is 38.15% in train versus 39.58% in test. For open stores, Sales–Customers correlation is 0.824. Mean sales rise from 5,929 without promo to 8,228 with promo, while mean customers rise from 697 to 844 with promo. December has the strongest monthly average sales.

Promotion distribution



Seasonality



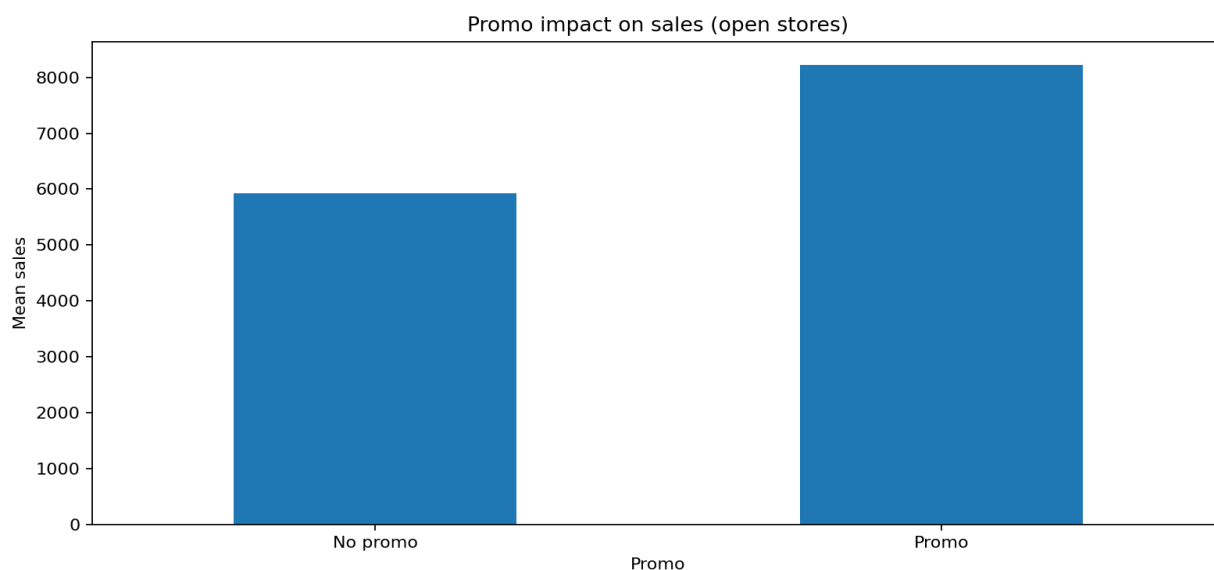
ML model and validation

The production model is LightGBM inside a sklearn Pipeline. The target is $\log_{1p}(\text{Sales})$, then predictions are returned to the original scale with `expm1`. Features include date parts, weekday/weekend, month position, cyclical encodings, competition age, Promo2 age/activity, holiday proximity, store metadata and Store×Promo interaction. Customers is excluded from sales modelling because it is not available in test.

Validation

July 2015 is held out as a chronological validation period. RMSLE = 0.0535; MAE = 256.24; RMSPE = 0.0573. RMSLE is the primary loss because it treats proportional errors more evenly for the skewed non-negative sales target.

Promotion effect

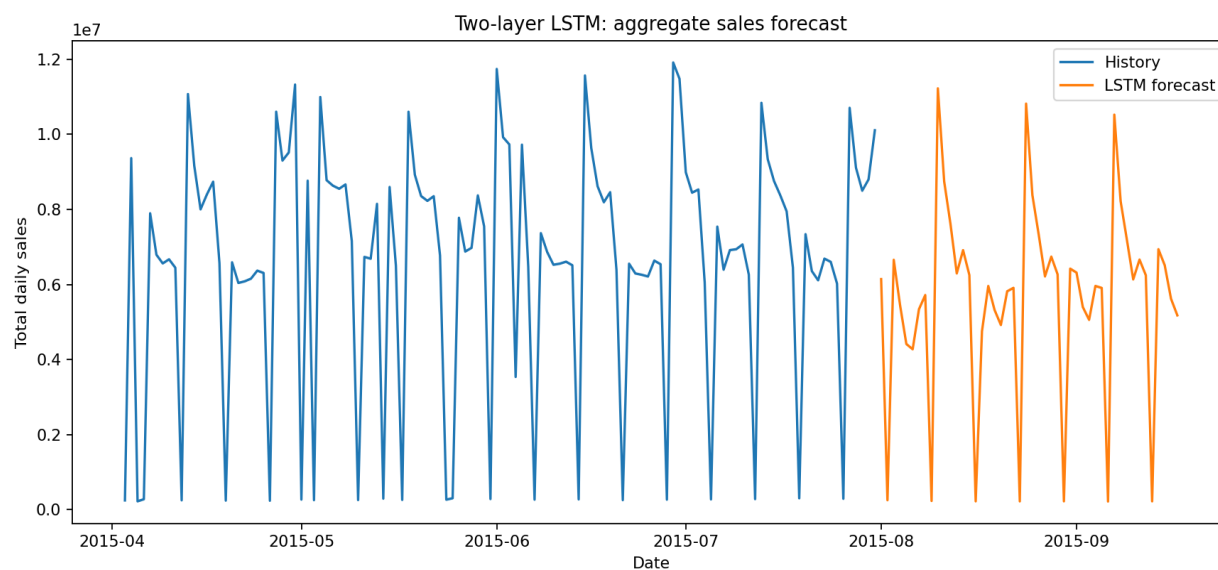
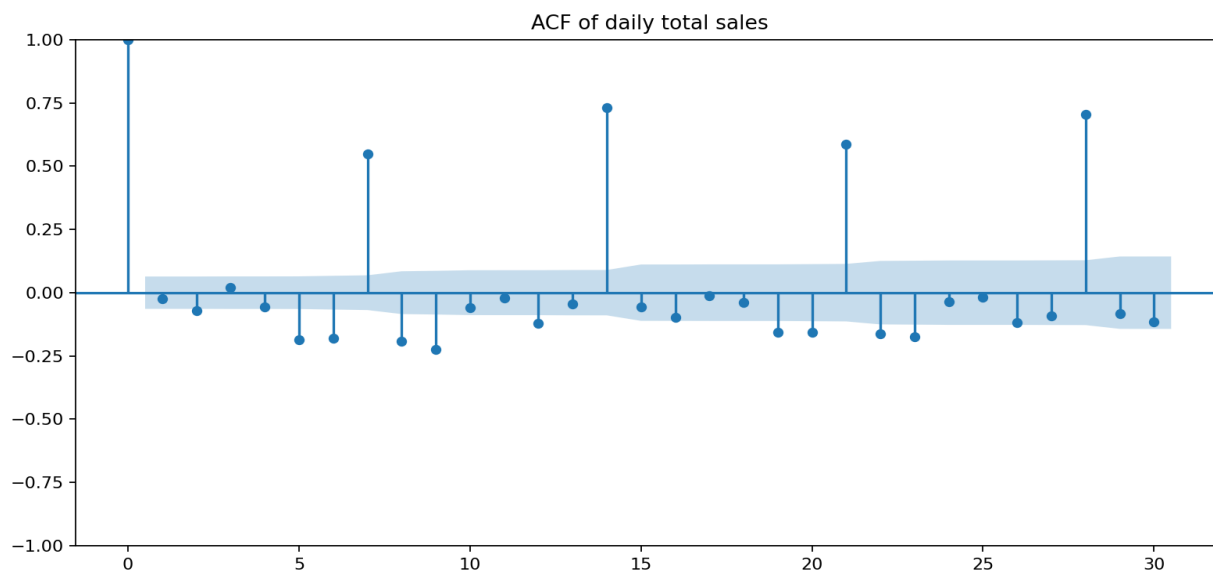


Sales vs customers



Deep Learning — two-layer LSTM

The aggregate daily Sales series was tested for stationarity. ADF p-value = $6.44e-05$, below 0.05, so differencing is not required. A 30-day sliding window is scaled to (-1,1), then a two-layer PyTorch LSTM predicts the next daily aggregate sales value. A log transform stabilizes variance. ACF/PACF and a 48-day demonstration forecast are included in outputs.



Deployment and MLOps

The Flask app loads serialized sales and customer models, accepts single-row inputs or CSV uploads, displays predicted sales/customers and a chart, and provides a downloadable CSV. MLflow tracking code creates multiple LightGBM runs with different num_leaves values and logs parameters, metrics and model artifacts. DVC configuration tracks the supplied data and model/submission outputs.

Submission checklist

EDA notebook and plots • sklearn pipeline • feature engineering • loss/metrics • feature importance • empirical prediction intervals • serialized ML models • two-layer LSTM • MLflow experiment script • DVC configuration • Flask dashboard • submission.csv • GitHub-ready structure.

Data-quality rule

The test set contains 11 missing Open values, all for Store 622. The production submission treats missing Open as 0 (closed), so those rows receive zero sales and zero customers. This rule is documented in the code and README.